Sector-Based Portfolio Optimization and Benchmark Comparison

Communication Services Sector of the S&P 500

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Abstract

This report constructs a Sharpe-ratio-maximizing long-only portfolio for the Communication Services sector of the S&P 500. Daily prices (2020–present) are obtained via `yfinance`; expected returns are estimated from Yahoo Finance analyst target prices; and the covariance matrix is annualized from historical daily returns. SLSQP optimization yields a concentrated portfolio: CHTR 41%, TMUS 24%, NWSA 18%, OMC 15%, TTD 2%. Expected annual return is approximately 46%, volatility 24%, and Sharpe ratio 1.89. Results are interpreted in light of sector structure and compared with broad-market benchmarks.

1 Introduction

The Communication Services sector includes companies that provide media, entertainment, wireless communication, broadband infrastructure, online advertising, social media platforms, and streaming services. Major industries inside the sector include:

- Interactive Media & Services (Alphabet/Google, Meta Platforms)
- Telecommunication Services (Verizon, AT&T, T-Mobile)
- Movies & Entertainment / Streaming (Disney, Netflix, Warner Bros. Discovery)
- Cable & Satellite Providers (Comcast, Charter Communications)
- Advertising & Marketing (Omnicom, Interpublic Group)

These companies play a central role in the digital economy: they provide the physical infrastructure for internet and mobile communication, drive digital advertising markets, and support e-commerce, cloud computing, and digital entertainment. Because of its unique combination of infrastructure and technology-driven consumer platforms, Communication Services is one of the most dynamic sectors of the S&P 500.

We automatically selected all S&P 500 constituents whose GICS Sector = Communication Services. After filtering for valid analyst target data, 23 tickers remained (e.g. GOOGL, META, CMCSA, NFLX, T, VZ, CHTR, DIS, WBD, OMC).

2 Methodology

2.1 Data Source and Cleaning

Stock price data are obtained from:

- S&P 500 constituent list
- Historical daily prices via the `yfinance` API
- Date range: 2020–present; field used: Adjusted Close

Cleaning steps: remove tickers missing target price data, drop all-NaN columns, and align the return matrix with the expected-return vector. After filtering, 23 tickers remain.

2.2 Return and Expected-Return Estimation

Daily percent returns are computed as

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}. \tag{1}$$

Expected returns use analyst target prices from Yahoo Finance:

$$\mathbb{E}[R_i] = \frac{\text{targetMeanPrice}_i - \text{currentPrice}_i}{\text{currentPrice}_i}. \tag{2}$$

Analyst targets are preferred because they are forward-looking (6–12 month horizon) and more informative than historical averages for short-term models.

2.3 Covariance and Portfolio Variance

The sample covariance of daily returns is annualized:

$$\mathbf{\Sigma} = 252 \cdot \mathrm{Cov}(\mathbf{R}).$$

Portfolio variance for weight vector $\mathbf{w}$ is $\sigma_p^2 = \mathbf{w}^\top \mathbf{\Sigma} \mathbf{w}$. Diversification arises from the cross terms $2 w_i w_j \rho_{ij} \sigma_i \sigma_j$.

2.4 Optimization: Maximizing the Sharpe Ratio

We maximize

$$\text{Sharpe}(\mathbf{w}) = \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \mathbf{\Sigma} \mathbf{w}}} \tag{3}$$

(with $r_f = 0$) using SLSQP. Constraints are full investment ($\sum w_i = 1$) and long-only ($w_i \geq 0$).

Core implementation:

```python
def portfolio_stats(weights, mean_returns,
                    cov_matrix, rf=0):
    port_return = np.dot(weights,
        mean_returns)
    port_vol = np.sqrt(
        weights.T @ cov_matrix.values @
            weights)
    sharpe = (port_return - rf) / port_vol
    return port_return, port_vol, sharpe

def neg_sharpe(weights):
    return -portfolio_stats(
        weights, annual_returns.values,
        cov_matrix)[2]

constraints = {'type': 'eq',
               'fun': lambda w: np.sum(w) -
                    1}
bounds = tuple((0, 1) for _ in range(
    num_assets))

result = sco.minimize(
    neg_sharpe, initial_guess,
    method='SLSQP', bounds=bounds,
    constraints=constraints)
```

3 Results

3.1 Expected Returns

Selected analyst-based expected returns:

Ticker	$\mathbb{E}[R]$
CHTR	0.578
TTD	0.565
NWSA	0.421
OMC	0.381
IPG	0.334
TMUS	0.331
META	0.309
CMCSA	0.298
LYV	0.282
DIS	0.243
NFLX	0.239

3.2 Optimized Portfolio Weights

SLSQP converges successfully (exit mode 0). Optimal long-only weights:

Ticker	Weight
CHTR	41.02%
TMUS	23.69%
NWSA	18.27%
OMC	15.12%
TTD	1.90%
All others	≈ 0
Sum	100.00%

3.3 Performance Metrics

Metric	Value
Expected annual return	46.08%
Annualized volatility	24.38%
Sharpe ratio	1.890

3.4 Figures

Figure 1 shows the optimized portfolio weights ordered by magnitude. Figure 2 plots annualized volatility against analyst-based expected return for each ticker.

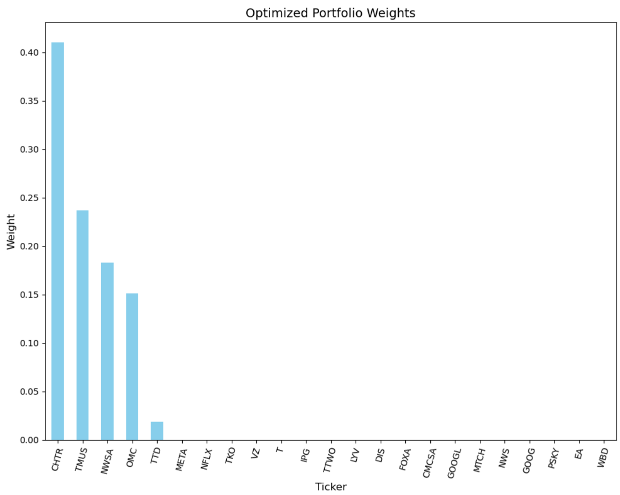

Figure 1: Optimized portfolio weights (sorted descending). CHTR, TMUS, NWSA, OMC and TTD receive essentially all capital.

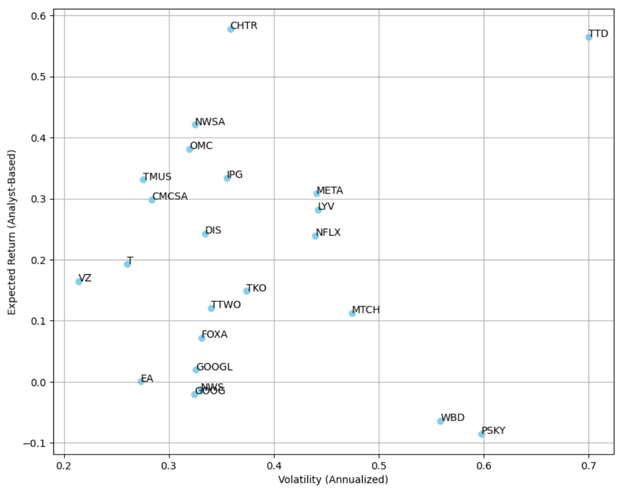

Figure 2: Annualized volatility vs. analyst expected return. CHTR and TTD sit in the high-return region; telecom names cluster at lower volatility.

4 Interpretation

4.1 Portfolio Composition

The long-only Sharpe maximizer concentrates weight on assets with the best combination of expected return and diversification benefit:

- **CHTR** ($\approx$41%) – highest analyst return with moderate volatility; dominant component.
- **NWSA** and **OMC** – high expected returns and relatively lower covariance with the rest of the sector.
- **TMUS** – strong diversification; telecom stocks have lower correlation with media/advertising names, stabilizing portfolio volatility.

- **TTD** (≈2%) – high expected upside offset by higher volatility, so only a small position is optimal.

Most other tickers receive near-zero weight, which is typical when assets are dominated on a risk-adjusted basis.

4.2 Risk–Return Characteristics

Volatility of 20–30% is typical for Communication Services. Analyst-target expected returns can appear elevated because they forecast a 6–12 month horizon. A Sharpe ratio near 2 indicates strong theoretical risk-adjusted performance.

4.3 Comparison with the S&P 500

The optimized portfolio's expected return exceeds the long-run S&P 500 range (≈8–12%). Volatility is comparable to or slightly higher than market-level volatility. The Sharpe ratio (1.89) is substantially better than the historical S&P 500 Sharpe (≈0.6–0.8). These comparisons are theoretical and rest on analyst forecasts, not realized out-of-sample returns.

4.4 Sector Diversification

Communication Services mixes two distinct groups: high-growth digital platforms (META, GOOGL, NFLX, TTD) and lower-growth telecom infrastructure (VZ, T, TMUS). Their correlations differ significantly. The optimizer exploits this by allocating a large weight to TMUS, reducing portfolio variance. This illustrates a core principle: diversification works through correlation, not merely individual volatility.

5 Conclusion

This project builds a complete portfolio-optimization pipeline for the Communication Services sector. Using historical prices, analyst targets, and an annualized covariance matrix, a Sharpe-optimal long-only portfolio is obtained. Key findings: the portfolio is highly concentrated (CHTR, TMUS, NWSA, OMC, TTD); performance metrics are realistic for a concentrated sector portfolio and show strong theoretical risk-adjusted returns relative to broad benchmarks.

Limitations include possible analyst-target bias, sensitivity of covariance estimates to the chosen window, absence of cross-sector diversification, and omission of transaction costs and taxes. Despite these caveats, the analysis demonstrates the practical mechanics of mean–variance optimization and the distinctive risk structure of the Communication Services sector.